

Learning outcomes

1. Identify and differentiate between public and private IP addresses.
2. Compare Azure VM sizes based on pricing, vCPU count, and RAM specifications.
3. Configure an Azure Virtual Machine with appropriate security settings.
4. Establish and manage inbound rules within Azure's network settings.
5. Utilize Wireshark to capture and analyze ICMP traffic.
6. Configure firewall settings to allow specific network traffic.
7. Establish Remote Desktop Protocol (RDP) connections with appropriate security considerations.
8. Set up and verify an IIS web server on an Azure VM.
9. Capture and interpret network traffic logs.
10. Document all configurations, findings, and troubleshooting steps.

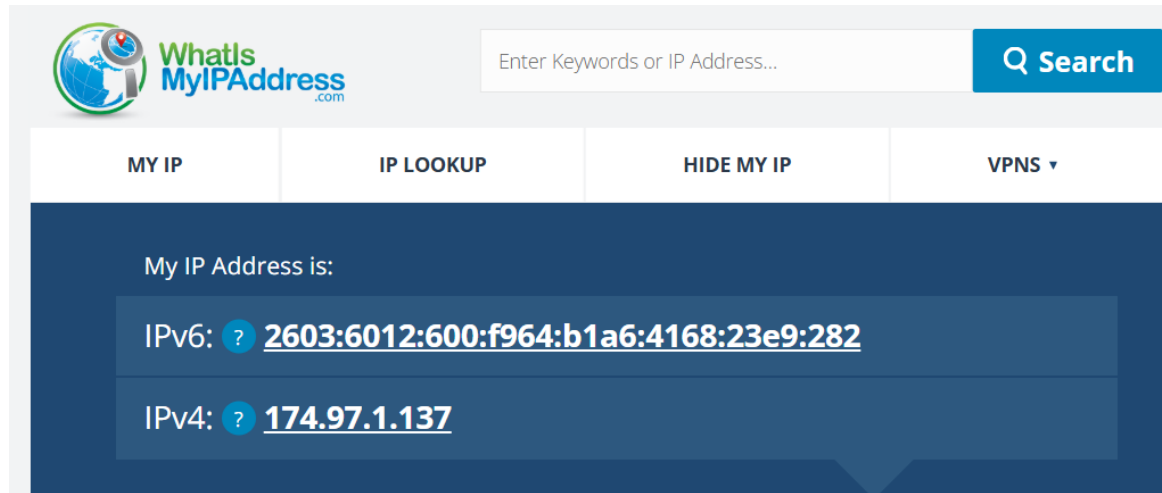
Terms and definitions

Term	Definition
Public IP Address	An IP address that is routable on the internet and assigned by an ISP.
Private IP Address	An IP address used within a private network, not routable on the internet.
Virtual Machine (VM)	A software-based emulation of a physical computer.
vCPU	A virtual CPU assigned to a VM for processing power.
RDP	Remote Desktop Protocol, used for remote access to computers.
Inbound Rule	A firewall rule that permits or blocks incoming traffic.
Wireshark	A network protocol analyzer used to capture and analyze packets.
ICMP	Internet Control Message Protocol, used for diagnostics such as ping requests.
Firewall	A network security system that monitors and controls incoming/outgoing traffic.
IIS	Internet Information Services, a web server for hosting websites on Windows.

Documentation

Follow the directions in the Lab Documentation Requirements

1. Determine the public IP address (i.e. not a private IP) that your PC is using and record that.



The screenshot shows the homepage of WhatIsMyIPAddress.com. At the top, there is a search bar with the placeholder text "Enter Keywords or IP Address..." and a blue "Search" button. Below the search bar, there are four navigation tabs: "MY IP", "IP LOOKUP", "HIDE MY IP", and "VPNS". The "MY IP" tab is selected. The main content area displays "My IP Address is:" followed by two lines of information: "IPv6: ? 2603:6012:600:f964:b1a6:4168:23e9:282" and "IPv4: ? 174.97.1.137".

My public IP address from my apartment today is 174.97.1.137.

2. For each of the VM sizes below: Determine the hourly pricing for each VM, the number of vCPUs and amount of RAM. Explain the differences in the models and the purpose they are designed for. Create a table to show your results

VM Size	vCPUs	RAM	Cost/hr	Differences
Standard B1	1	1GB	\$0.01	"Ideal for workloads that do not need continuous full CPU performance" Balanced CPU-to-memory ratio, ideal for testing and development, small to medium databases, and low to medium traffic web servers
Standard D2s_v3	2	8GB	\$0.10	"General purpose" Balanced CPU-to-memory ratio, ideal for testing and development, small to medium databases, and low to medium traffic web servers; more robust than the B1 size.
Standard E4s_v3	4	32GB	\$0.25	"Memory optimized" High Memory-to-CPU ratio great for relational database servers, medium to large caches, and in-memory analytics.
Standard F8s_v2	8	16GB	\$0.34	"Compute optimized" High CPU-to-memory ratio, good for medium traffic web servers, network appliances, batch processes, and application servers.

Source: <https://learn.microsoft.com/en-us/azure/virtual-machines/sizes/overview?tabs=breakdownseries%2Cgeneralizelist%2Ccomputesizelist%2Cmemorysizelist%2Cstoragesizelist%2Ccpusizelist%2Cfpgasizelist%2Chpcsizelist>

3. Create an Azure VM. You pick the size.
 - a. Set the Inbound port rules for Public ports to be “none”.
 - b. Explain what this will mean in terms of using your VM

✓ Your deployment is complete

 Deployment name: CreateVm-MicrosoftWindowsServer.WindowsSe... Start time: 1/26/2025, 9:24:15 PM
Subscription: [Azure for Students](#) Correlation ID: 694e986e-3b52-452f-b886-7a645c7195bd 
Resource group: [Labs](#)

Deployment details

Resource	Type	Status	Operation details
✓ Lab2VM1	Microsoft.Compute/virtualMachi...	OK	Operation details
✓ lab2vm1873_z2	Microsoft.Network/networkInter...	Created	Operation details
✓ Lab2VM1-nsg	Microsoft.Network/networkSecu...	OK	Operation details

This is a screenshot of the created VM – Lab2VM1.

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ ☒ None
☐ Allow selected ports

Setting the Inbound port rules for Public ports to be ‘none’ means there are no network ports accessible from the public internet, shown by the above screenshot.

4. In Azure, go to the Network Settings (i.e. Firewall) and Create port rule, then select Inbound port rule
 - a. Add a TCP rule to allow RDP (Remote Desktop Protocol) from the public IP address you are using.
 - i. The ATLC-339 lab 216.196.128.232/29. If you are performing the lab remotely add your public IP address
 - b. Verify that you are able to RDP into your VM from the lab
 - c. Explain why you get the message “The identify of the remote computer cannot be verified...”
 - d. Explain what simple user step is required to prevent this message from being displayed for any user. The “Don’t ask me again” is not the solution.

Network security group **Lab2VM1-nsg** (attached to networkInterface: **lab2vm1873_z2**)
Impacts 0 subnets, 1 network interfaces

[+ Create port rule](#)

Search rules	Source == all	Destination == all	Protocol == all	Action == all	
Priority ↑	Name	Port	Protocol	Source	Destination
Inbound port rules (4)					
100	AllowMyIpAddressRDPInbound	3389	TCP	174.97.1.137	Any

The TCP Rule allowing RDP from my IP is shown above.

Name ↑↓	Subscription ↑↓	Resource group ↑↓	Location ↑↓	Status ↑↓	Operating system ↑↓	Size ↑↓	Public IP address ↑↓
☐ Lab2VM1	Azure for Students	Labs	East US	Running	Windows	Standard_DS1_v2	172.190.50.245



A running instance of Lab2VM1, confirmed by the matching IP addresses and the VM instance RDP connection is shown above.

The identity of the remote computer cannot be verified because it has a self-signed certificate that has not been authenticated by any official source. Since the instance was just created, it has not obtained a certificate from a trusted certifying authority. This poses a security risk if it's not a trusted machine, so the message pops up as an added level of security.

To prevent this message from being displayed to any user, one can install the certificate in the Trusted Root Certification Authorities store.

- Using a web browser, login into <https://mydesktop.cincinnatistate.edu> Use your College credentials.

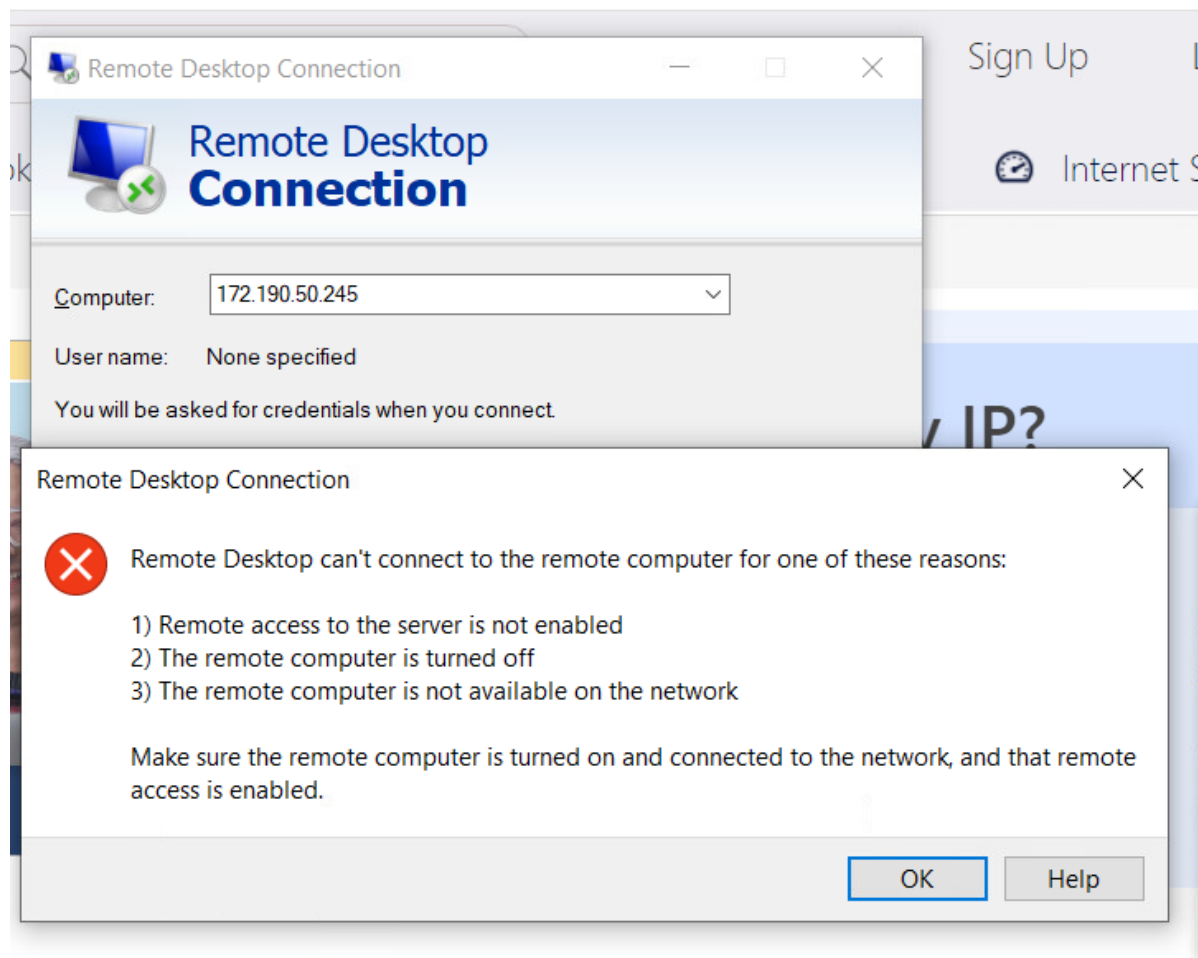
Note: The College's remote desktop uses VMware's Virtual Desktop Infrastructure (VDI)

- Determine the public IP that your VDI session is using
- Open Remote Desktop Connection (RDC) and try to login to your VM
- Capture the message RDC returns
- In the Network Settings, add another rule to allow RDP from the College's IP block 199.218.107.0/24
- Verify that you can use RDC to connect to your VM

What Is My IP?

My Public IPv4: [199.218.107.3](#) 

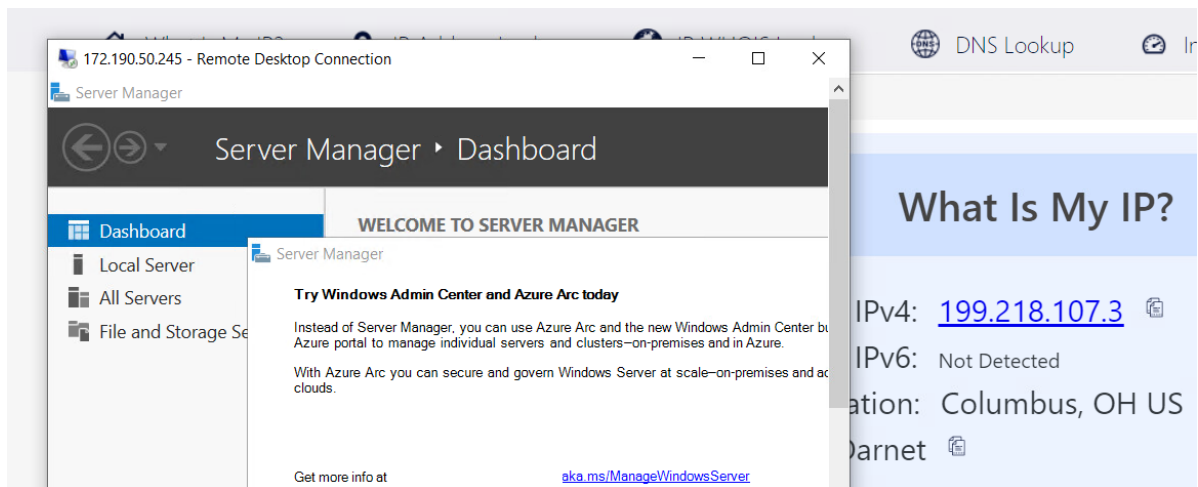
The VDI's public IP is 199.218.107.3



The RDP failed from the VDI because there is no rule in the VM to allow its access.

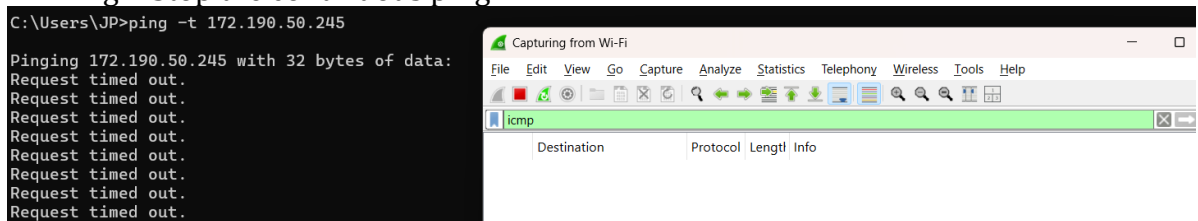
Inbound port rules (5)						
100	AllowMyIpAddressRDPInbound	3389	TCP	174.97.1.137	Any	
110	AllowCidrBlockRDPInbound	3389	TCP	199.218.107.3	Any	
65000	AllowVnetInBound ⓘ	Any	Any	VirtualNetwork	VirtualNetv	

New Rule added to allow the VDI IP to access the VM via RDP

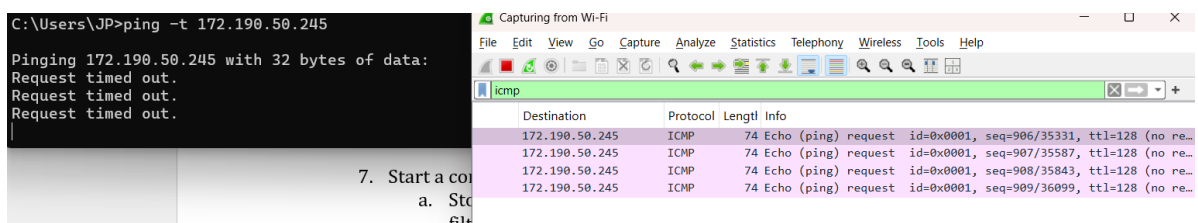


Verified connection from the VDI

6. Start a continuous ping from your desktop to your VM.
 - a. Apply the ICMP filter to Wireshark. Verify that you don't see any ICMP traffic by starting a continuous ping to your VM. Make sure Windows Defender Firewall is turned on.
 - b. Change the Windows Defender Firewall to allow IPv4 ICMP inbound.
 - c. Add a firewall rule to your VM to allow ICMP from your public IP. Once you "Add" the rule, start a timer (use smart phone, etc.)
 - d. Record the number of seconds before you see successful pings to your VM
 - e. Verify that you see ICMP Echo Request and Echo Reply
 - f. Using the Wireshark data, determine the elapsed time in mS. Show your result as X.XX mS format. Repeat this for 3 sets of Echo Request/Reply packets. Include the screenshots and your calculations
 - g. Stop the continuous ping



A continuous ping with nothing on WireShark. There is no ICMP traffic passing through to the VM.



7. Start a continuous ping from your desktop to your VM.
 - a. Stop the continuous ping

Now we are capturing the traffic going out, but nothing is echoing back.

Request timed out.	72	5.190240	192.168.1.96	172.190.50.245	ICMP	74 Echo (ping) request
Request timed out.	371	10.177368	192.168.1.96	172.190.50.245	ICMP	74 Echo (ping) request
Request timed out.	627	15.164182	192.168.1.96	172.190.50.245	ICMP	74 Echo (ping) request
Request timed out.	845	20.165891	192.168.1.96	172.190.50.245	ICMP	74 Echo (ping) request
Request timed out.	970	25.177424	192.168.1.96	172.190.50.245	ICMP	74 Echo (ping) request
Reply from 172.190.50.245: bytes=32 time=36ms TTL=108	971	25.213569	172.190.50.245	192.168.1.96	ICMP	74 Echo (ping) reply
Reply from 172.190.50.245: bytes=32 time=35ms TTL=108	1054	26.198490	192.168.1.96	172.190.50.245	ICMP	74 Echo (ping) request
Reply from 172.190.50.245: bytes=32 time=34ms TTL=108	1055	26.225944	172.190.50.245	192.168.1.96	ICMP	74 Echo (ping) reply
Reply from 172.190.50.245: bytes=32 time=35ms TTL=108	1067	27.207744	192.168.1.96	172.190.50.245	ICMP	74 Echo (ping) request
Reply from 172.190.50.245: bytes=32 time=36ms TTL=108	1074	27.242454	172.190.50.245	192.168.1.96	ICMP	74 Echo (ping) reply

Here are the side by side command prompt and the wireshark capture. You can see that the timed out requests are shown on the right as requests without replies. The First reply for the ping is sen in the packet #971.

It took 8.23 seconds to receive the first reply according to my stopwatch.

1216	31.270113	192.168.1.96	172.190.50.245	ICMP	74 Echo (ping) request
1217	31.305031	172.190.50.245	192.168.1.96	ICMP	74 Echo (ping) reply
1224	32.284148	192.168.1.96	172.190.50.245	ICMP	74 Echo (ping) request
1225	32.319815	172.190.50.245	192.168.1.96	ICMP	74 Echo (ping) reply
1226	33.303107	192.168.1.96	172.190.50.245	ICMP	74 Echo (ping) request
1227	33.339809	172.190.50.245	192.168.1.96	ICMP	74 Echo (ping) reply

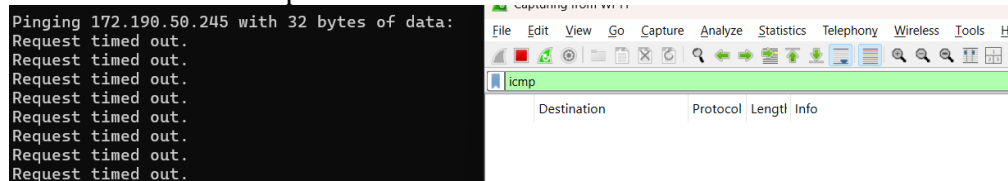
Here are three request/reply pairs of ping in Wireshark used for the calculations below.

The time that packet 1216's response took to get back is
 $31.305031 - 31.270113 = 0.034918$ seconds, or 34.92 milliseconds..

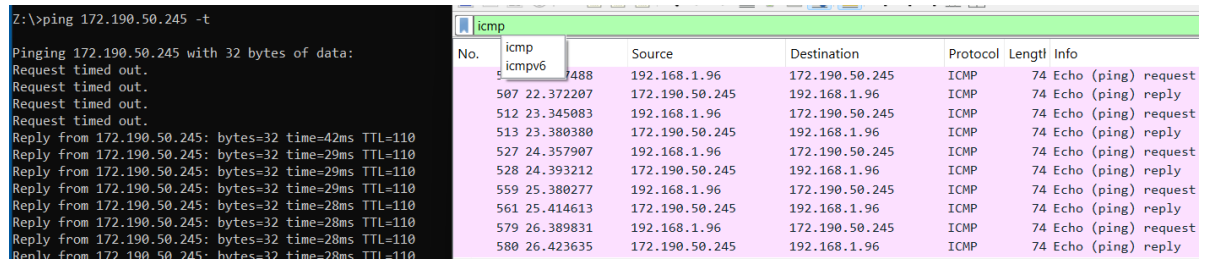
The time that packet 1224's response took to get back is
 $32.319815 - 32.284148 = 0.035667$ seconds, or 35.67 milliseconds.

The time that packet 1226's response took to get back is
 $33.339809 - 33.303107 = 0.036702$ seconds, or 36.70 milliseconds.

7. Start a continuous ping from your VDI to your VM
 - a. Stop, and then restart your Wireshark capture. Make sure your ICMP filter is applied.
 - b. Verify that you do not see any ICMP traffic.
 - c. Add a firewall rule to allow ICMP from the College's VDI
 - d. Examine the results of both your continuous ping and ICMP capture in Wireshark. Explain your results. What conclusion can you draw from this data? Explain



Once again, a continuous ping without any traffic detected above.



The image shows a Windows command prompt on the left and a Wireshark packet capture window on the right. The command prompt shows a continuous ping to 172.190.50.245. The first three pings time out, and then several successful replies are shown with response times around 28-42ms. The Wireshark window shows the ICMP traffic corresponding to these pings, with a filter set to 'icmp'. The packet list shows alternating request and reply packets between 192.168.1.96 and 172.190.50.245.

No.	icmp	Source	Destination	Protocol	Length	Info
507	icmpv6	192.168.1.96	172.190.50.245	ICMP	74	Echo (ping) request
512	icmpv6	172.190.50.245	192.168.1.96	ICMP	74	Echo (ping) reply
513	icmpv6	192.168.1.96	172.190.50.245	ICMP	74	Echo (ping) request
513	icmpv6	172.190.50.245	192.168.1.96	ICMP	74	Echo (ping) reply
527	icmpv6	192.168.1.96	172.190.50.245	ICMP	74	Echo (ping) request
528	icmpv6	172.190.50.245	192.168.1.96	ICMP	74	Echo (ping) reply
559	icmpv6	192.168.1.96	172.190.50.245	ICMP	74	Echo (ping) request
561	icmpv6	172.190.50.245	192.168.1.96	ICMP	74	Echo (ping) reply
579	icmpv6	192.168.1.96	172.190.50.245	ICMP	74	Echo (ping) request
580	icmpv6	172.190.50.245	192.168.1.96	ICMP	74	Echo (ping) reply

After adding the rule, we can see the ping traffic go through again.

It took less time to get a signal for the VDI than for my laptop. It took only 2.16 seconds since adding the rule to get a ping reply instead of 8 seconds before.

Also, the VDI seemed to be quicker to get responses, getting them in 29 or 28 milliseconds instead of the earlier 35 or 36 milliseconds.

The conclusion I can draw is that there must be a faster path (maybe with less hops) between the VDI and the VM than between my laptop and the VM.

8. Using Server Manager, install Internet Information Server (IIS). Take all the defaults.
 - a. This will create a directory c:\inetpub\wwwroot. There is a default web page called iisstart.htm. Remove that web page,
 - b. Use AI or hand write the HTML for a simple webpage that shows your name and the current date. Create a text document (i.e. use Notepad) and past the HTML in the document and save it with the name and extension index.htm Then copy that file into wwwroot
 - c. First verify that you can use a web browser to reach the webpage from a web browser within the VM. Then verify you can reach the webpage from your public IP address.

ROLES AND SERVER GROUPS

Roles: 2 | Server groups: 1 | Servers total: 1

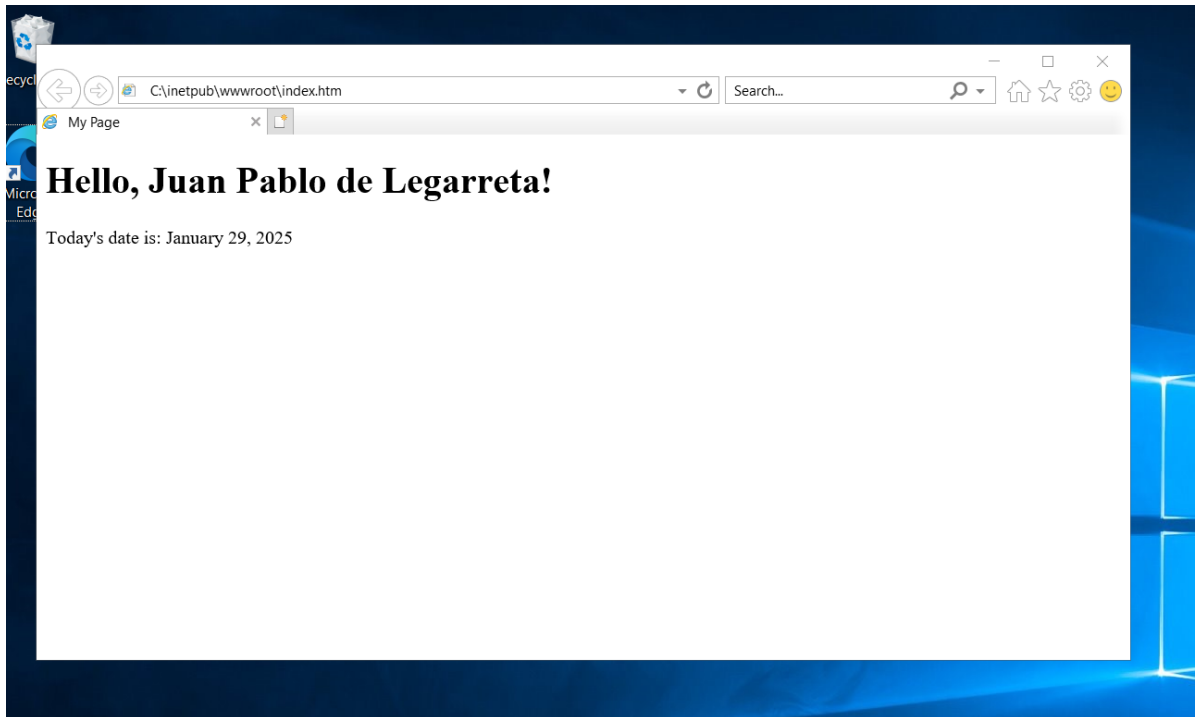
 File and Storage Services 1	 IIS 1
 Manageability	 Manageability
Events	Events
Performance	Services
BPA results	Performance
	BPA results

IIS Installed!

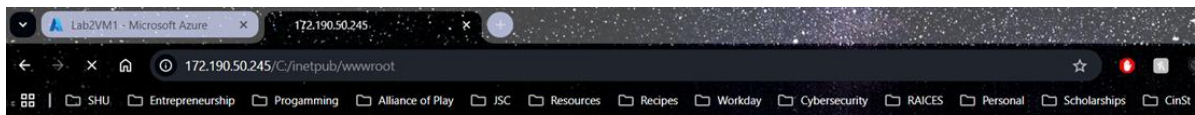
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Page</title>
  <script>
    function getCurrentDate() {
      const today = new Date();
      const options = { year: 'numeric', month: 'long', day:
'numeric' };
      return today.toLocaleDateString(undefined, options);
    }
  </script>
</head>
<body>
  <h1>Hello, Juan Pablo de Legarreta!</h1>
  <p>Today's date is: <span id="current-date"></span></p>

  <script>
    document.getElementById('current-date').textContent =
getCurrentDate();
  </script>
</body>
</html>
```

Index.htm is written!



Accessible on the VM!



Hello, Juan Pablo de Legarreta!

Today's date is: January 29, 2025

And accessible via the internet!

Consider creating an image of your VM for future labs.

9. Delete your VM. Verify that your VM has been deleted. Failure to do so can quickly deplete your credits.

Questions:

Q1. Research and explain a use case that aligns with each of the Azure VM models in #2 above.

Source:

<https://learn.microsoft.com/en-us/azure/virtual-machines/sizes/overview?tabs=breakdownseries%2Cgeneralizelist%2Ccomputesize>

[list%2Cmemorysizelist%2Cstoragesizelist%2Ccpusizelist%2Cfpgasizelist%2Chpcsize
list](#)

Standard B1

“Ideal for workloads that do not need continuous full CPU performance”

Balanced CPU-to-memory ratio, ideal for testing and development, small to medium databases, and low to medium traffic web servers.

The B1 has a ‘burstable’ feature that allows for CPU performance to ‘burst’ up to the maximum on an as-needed basis, but idles at a lower CPU performance to reduce cost. A proof of concept application would be a good use case for the B1 VM, as it could be tested without too much cost.

Standard D2s_v3

“General purpose”

Balanced CPU-to-memory ratio, ideal for testing and development, small to medium databases, and low to medium traffic web servers; more robust than the B1 size.

Because the D2s_v3 has consistent CPU usage, the use case for this type of VM would be more akin to testing and development of web services or hosting a small to medium database. It does not have the computing power to efficiently host a large database or a high traffic server.

Standard E4s_v3

“Memory optimized” High Memory-to-CPU ratio great for relational database servers, medium to large caches, and in-memory analytics.

The E4s_v3’s high Memory-to-CPU ratio is great for analytics that are done in memory as well as running relational database servers. Relational database servers strategically cache data in RAM to process it faster, and thus greatly benefits from a high Memory-to-CPU ratio. This model also helps keep this kind of VM affordable, as it favors Memory over CPU instead of having a high balanced configuration that would cost more money.

Standard F8s_v2

“Compute optimized” High CPU-to-memory ratio, good for medium traffic web servers, network appliances, batch processes, and application servers.

The F8s_v2 has high CPU-to-Memory ratio, making it perfect for application servers that require more computing power than databases. It also is the reverse of the E4s_v3 in that it has an imbalance of CPU to Memory, favoring the CPU to complete these complex calculations while keeping the VM affordable.

Q2. Why does Windows block ICMP traffic by default?

Windows blocks ICMP traffic by default as a defensive measure to boost security. ICMP traffic can be used to flood a network and create a Denial of Service (DOS) attack. Automatically stopping it at the firewall helps to keep systems secure from this kind of attack.

Source: <https://blog.securityevaluators.com/icmp-the-good-the-bad-and-the-ugly-130413e56030>

Q3. What network address and CIDR mask should be entered if you want to allow access to your VM from the entire class C subnet that the VDI is in?

199.218.107.0/24

Q4. From an IT support standpoint research and explain that value of using VDI and provide two use cases. Explain how VDI is different than using RDP with Windows Server Remote Desktop Services.

Using a Virtual Desktop Interface provides a secure configuration of a system with pre-set applications hosted on a cloud and not tied to a specific endpoint. This can be extremely useful for organizations with large numbers of remote workers who need to log in from different places. Another use case for using a VDI is that it can handle managing multiple instances simultaneously, while RDP can only access one desktop at a time.

This is why VDI can be more useful to an IT support group, since they can handle multiple problems from the same device using this protocol, making it more flexible than RDP. VDI is also more flexible than RDP in that it can handle a multitude of multiple operating systems and applications, whereas RDP will be at the mercy of the endpoint that the user is connecting to.

Source: <https://cloudzy.com/blog/vdi-vs-rdp-vs-rds/>

Q5. When showing ping round trip times, explain what is meant by 3 significant digits. Provide examples to illustrate this

Ping round trip times are measured in milliseconds, or .xxx seconds. If a ping takes .123 seconds, it is said to take 123 milliseconds. Usually pings return somewhere between 0 and 150 milliseconds. The amount of milliseconds is rounded by the three significant digits. In other words, the fraction of one second (the numbers after the decimal point) are significant to three digits. The third digit in the decimal is rounded up or down based on what the fourth digit is. For example, if the time is calculated to be .0248 seconds, the round trip time will be displayed as 25 milliseconds, since the 8 causes the 4 to round up. However, if the time had been .0243 seconds, the round trip time would be 24 milliseconds, as the 3 causes the 4 to 'round down' or stay the same.

```
C:\Users\JP>ping 8.8.8.8
```

```
Pinging 8.8.8.8 with 32 bytes of data:
```

```
Reply from 8.8.8.8: bytes=32 time=16ms TTL=57
```

```
Reply from 8.8.8.8: bytes=32 time=17ms TTL=57
```

```
Reply from 8.8.8.8: bytes=32 time=33ms TTL=57
```

```
Reply from 8.8.8.8: bytes=32 time=16ms TTL=57
```

```
Ping statistics for 8.8.8.8:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 16ms, Maximum = 33ms, Average = 20ms
```